

Executive Summary: Loan Default Prediction Model (1 to 3 pages)

If I were a bank executive, what would I need to approve or use this model?

- Visuals (strongly encouraged)
- No code
- No hyperparameters
- No jargon
 - Model architecture details (e.g., “we used XGBoost with...”)
 - Grid search/tuning discussion
 - Data cleaning steps
 - Long metric tables

1. Objective

What problem are we solving?

Why does it matter?

Example

We developed a machine learning model to predict the likelihood that a borrower will default on a loan at the time of application. The goal is to improve credit risk assessment, reduce financial losses, and support more informed lending decisions.

2. Key Results

Model performance (in plain English).

Business-relevant metrics

- Avoid accuracy alone.
- Prefer:
 - % of defaults correctly identified (recall)
 - Precision for high-risk flags
 - Lift/improvement over current process

Example

The model identifies 78% of likely defaults

Among flagged high-risk applicants, 65% default.

Using the model reduces expected losses by ~22%.

3. How It Would Be Used

Executives want to know: *what do we do with this?*

Explain:

- Where it fits in the decision process
- What actions it triggers

Example

- Each loan applicant is assigned a risk score at submission
- Applicants in the top 15% risk group are:
 - declined, or
 - sent for manual review
- Medium-risk applicants may receive adjusted interest rates or loan terms
- Low-risk applicants proceed through standard approval

4. Key Drivers of Default

Include:

- Top 3–5 features influencing default
- Explain in business terms

Example

The model identifies several strong predictors of default:

- Debt-to-income ratio: Higher ratios significantly increase risk
- Past delinquencies: Borrowers with prior missed payments are much more likely to default
- Credit history length: Shorter histories are associated with higher risk
- Loan amount relative to income: Larger loans relative to income increase default probability

5. Business Impact

Translate model performance → money/outcomes.

Example

- Estimated reduction in default-related losses: 15–25% annually
- Improved risk segmentation enables:
 - more accurate pricing of loans
 - fewer high-risk approvals
 - increased confidence in approving low-risk borrowers

6. Risks and Limitations

Include:

- Data limitations
- Model uncertainty
- Potential bias/fairness concerns

Example

- Model performance depends on historical data and may change with economic conditions
- Potential bias may exist if historical lending decisions were biased
- Requires ongoing monitoring and periodic retraining to maintain accuracy

7. Recommendation

End with a decision.

Example

We recommend a 3-month pilot deployment alongside the current underwriting process to validate real-world performance. Based on pilot results, the model can then be integrated into the standard loan approval workflow.